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MODULE- I

Alternators –

Basic principle, construction features, types of armature winding, EMF equation

INTRODUCTION

- A Synchronous machine is an AC machine in which the rotor moves at a speed which bears a constant relationship to frequency of currents in the armature (stator) winding
- Synchronous machines are generally constructed in larger sizes as small size alternators are not economical

Classification of Synchronous machines

According to application, may be classified as

1. *Synchronous generators or Alternators* (converts Mechanical energy from a prime mover to Electrical energy)
2. *Synchronous Motors* (converts Electrical energy into Mechanical energy)
3. *Synchronous compensators/condensors* (operates at no load and shaft is not connected to any prime mover or mechanical load and is used to control reactive power in power supply)

Alternator - Principle of operation

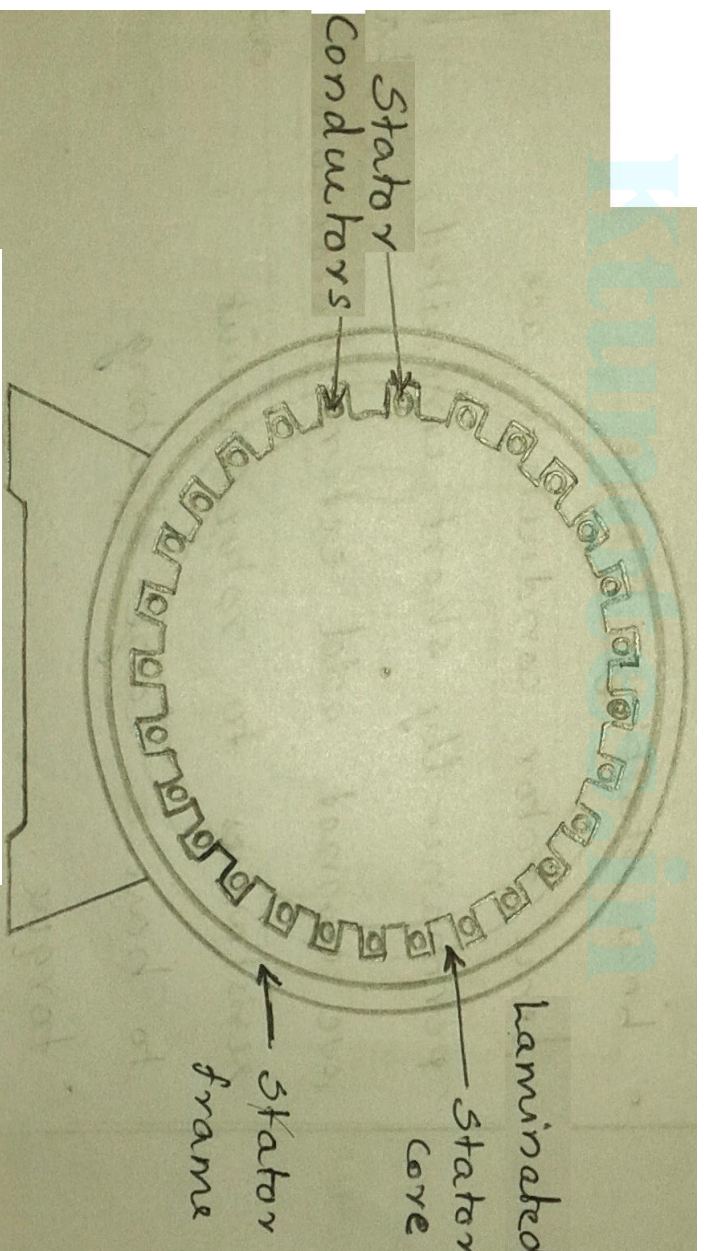
- Principle of operation is fundamentally same as that of a DC generator (Electromagnetic induction, Faradays laws)
- In an Alternator, EMF induced due to relative motion between conductor & magnetic field
- Here no need of converting induced AC EMF in armature to DC
- **So commutator is not needed as in a DC generator**
- Alternators are driven at very definite speed as the frequency of generated EMF is determined by that speed.
- Because of the absence of commutator, they are simple & possess several advantages over DC generator.

Construction

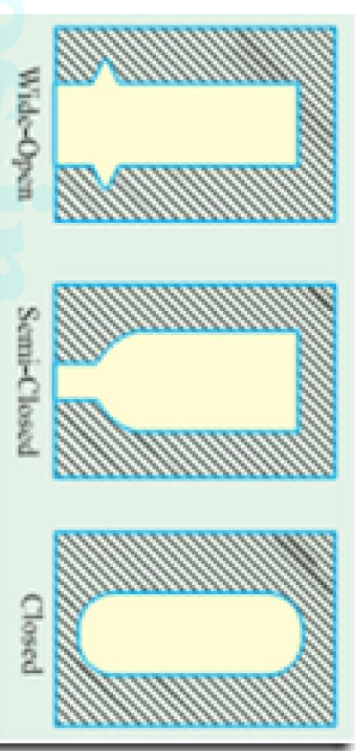
- Mainly two parts, namely armature (stator) and field magnet system (rotor)

a) Armature (stator)

- Main parts of stator are *stator core*, *stator winding* & *stator frame*
- *Stator core* contain an iron ring made of silicon steel laminations



- Slots are cut on the inner periphery of the stator core, in which stator conductors are placed.
- Stator core is laminated to reduce eddy current loss.
- It also contain axial & radial ventilating ducts to provide efficient cooling
- Slots can be open, semi closed or totally closed
- Open slots - are commonly used because coils can be insulated before placing in the slots and the expenditure is low. Also replacement of defective coil is easy here.
- But it got disadvantage of distributing the air gap flux into branches which tend to produce ripples in EMF wave
- Semi closed slots - it overcomes the drawback of open slots, but coils cannot be insulated before placing in the slots
- Totally closed slots - are rarely used



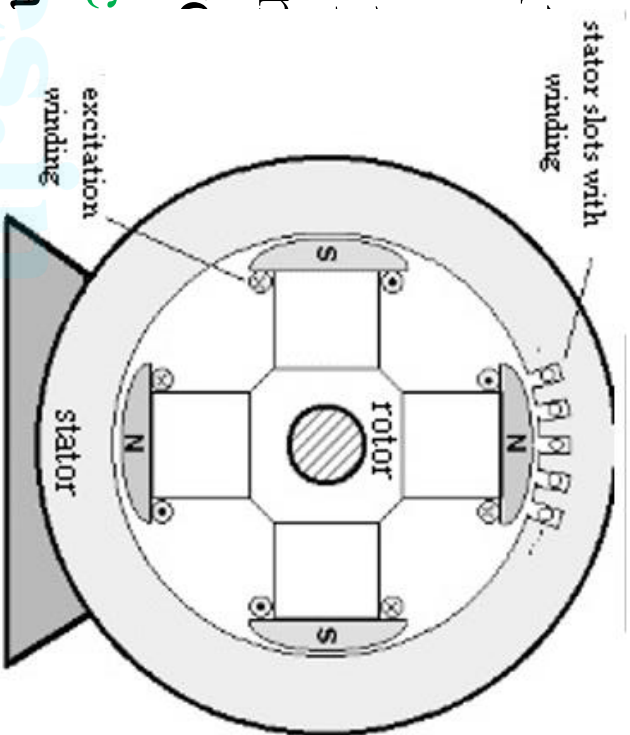
- Stator contain a 3 phase winding which is normally star connected
- Stator conductors are made of copper or aluminium
- Windings can be single layer or double layer, lap or wave, concentrated or distributed
- The whole structure is held in a frame made of cast iron or welded steel plate

b) Field magnet system (Rotor)

- The field magnets produce the magnetic field required for the generation of AC voltage.
- Field magnets are supplied with DC voltage.
- For this external DC sources are used.
- If an external DC source is used, exciting current is supplied to rotor through slip rings & brushes.
- In order to make the alternator self contained & independent of other sources, usually a small DC generator (Exciter generator) is installed & coupled to the shaft of alternator.
- Recently brushless excitation system is used, in which a small 3 phase alternator is used as exciter & a group of rectifiers will convert AC to DC & will supply to field system.
- Hence brushes & slip rings are eliminated.
- There are mainly 2 types of rotor construction

1) Salient pole type

- Used with slow & moderate speed al
- Less expensive.
- This type of machine is not used for to the difficulty in obtaining sufficient
- Also it cause large windage loss & pr
- Like in DC machine, **it contain projec** attached on the outer periphery of sha appearance.



- Field poles contains laminated pole core & pole shoe (to reduce eddy current losses) and field windings
- The flywheel type rotor is constructed to give adequate mechanical strength to with stand centrifugal & driving forces. At the same time it provides necessary path for the magnetic fluxes

- Damper windings (short circuited copper bars) are provided on rotor to damp rotor oscillations
- Since the salient pole machine is a low speed machine, to produce desired frequency AC voltage, *it is provided with large number of poles (4 to 60)* (synchronous speed, $N_s = 120f/P$)
- To accommodate these large number of poles, these machines *have large diameter & small length*
- Pole shoes cover about 2/3 of pole pitch
- *Usually employed with hydraulic turbines*

2. Cylindrical or non-salient pole

It is used with high speed alternators

To reduce peripheral velocity of r reduced & axial length is increased.

Usually it has 2 poles in rotor

Field windings are placed in slots

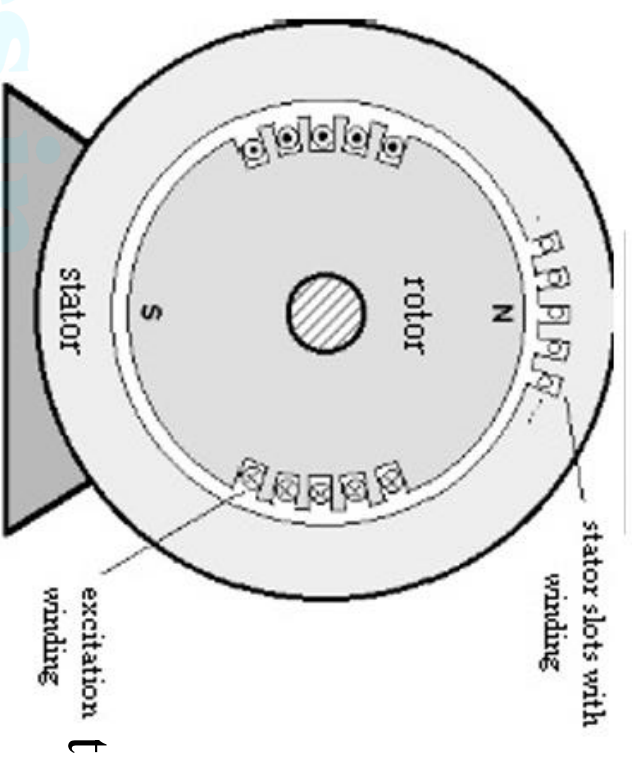
Normally $2/3^{\text{rd}}$ of rotor is slotted for the slots to form the pole faces

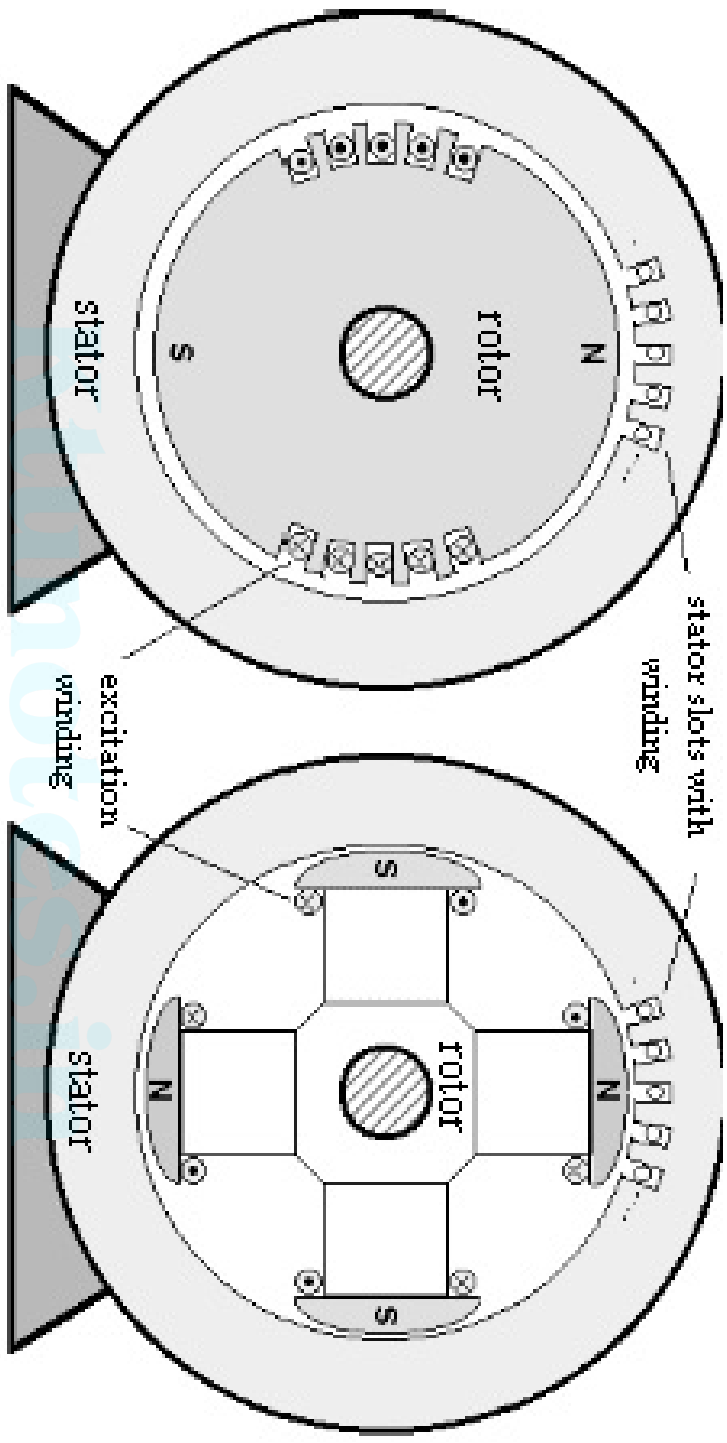
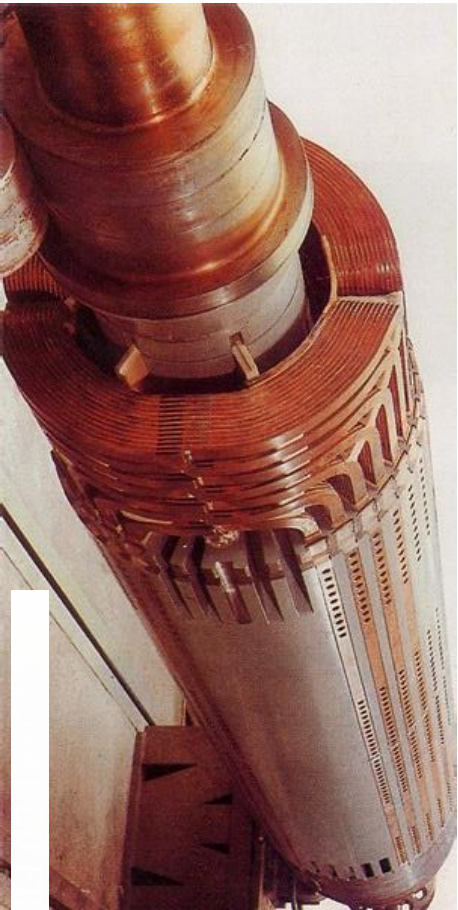
It has robust construction & noise less operation

Less windage loss

No need of damper windings

Nearly sinusoidal flux distribution around the periphery to give a better EMF waveform





Classification of Synchronous machines.

Based on *the number of phases*, they are classified as

- i. Single phase – All the armature coils are connected for addition of individual voltages and a single AC output
- ii. Two phase – Armature has two set of windings & two output voltages are 90° out of phase
- iii. Three phase – armature has three set of windings & three output voltages are 120° out of phase

Based on construction, they are classified into

i. Rotating armature type

- It looks like a DC generator except that there are 3 slip rings in place of commutator
- The required magnetic field is produced by DC electromagnets placed in Stator
- The current generated in rotor is collected by brushes & sliprings
- This arrangement is economical for low voltage generators with ratings upto 250k VA.

ii) Rotating field type

- Here the required magnetic field is produced by DC electromagnets placed in rotor
- DC electromagnets are powered either by brushless excitation system or from an external DC source through slip rings & brushes
- The electric power generated in stator can be taken out directly
- All medium & large sized machines are constructed in this manner

❖ The advantages of stationary armature & revolving field system

- *It is easier to insulate stationary armature winding for high voltage (as high as 33kV)*
- *The load circuit can be connected directly with fixed terminals of stator without passing through slip rings & brushes. In rotating armature type, load circuit is required to be connected to armature through slip ring & brush*
- *Armature winding can be more easily braced in a rigid frame to prevent any deformation.*
- *Armature winding is cooled readily by providing many air passages & cooling ducts in stator frame.*
- *Only two slip rings are required for the supply of DC voltage to the rotor*
- *The excitation current is relatively small. So slip rings & brush need light construction*
- *Due to simple, light & robust construction of rotor, higher speeds of operation is possible.*

Speed of rotation of an Alternator

- Synchronous generators are by definition synchronous, meaning that the frequency of induced emf in stator is locked in or synchronized with the mechanical rate of rotation of the rotor

$$\text{i.e, } N_s = \frac{120f}{P}$$

where, N_s = Speed of rotor in rpm (Synchronous speed)

f = stator current frequency in Hz

P = Number of poles

Armature (stator) windings

- **Armature windings of alternators** are different from that of d.c machines.
- Basically, three phase **alternators** carry three sets of windings arranged in the slots in such a way that there exists a phase difference of 120° electrical between the induced e.m.f.s in them.
- In a dc machine, winding is closed while in **alternators** winding is open i.e., two ends of each set of the winding are brought out.
- In three phase alternators, the six terminals are brought out which are finally connected in star or delta and then the three terminals are brought out.
- Each set of windings represents winding per phase and induced emf in each set is called induced emf per phase denoted as E_{ph} .

Types of Armature Windings in Alternator:

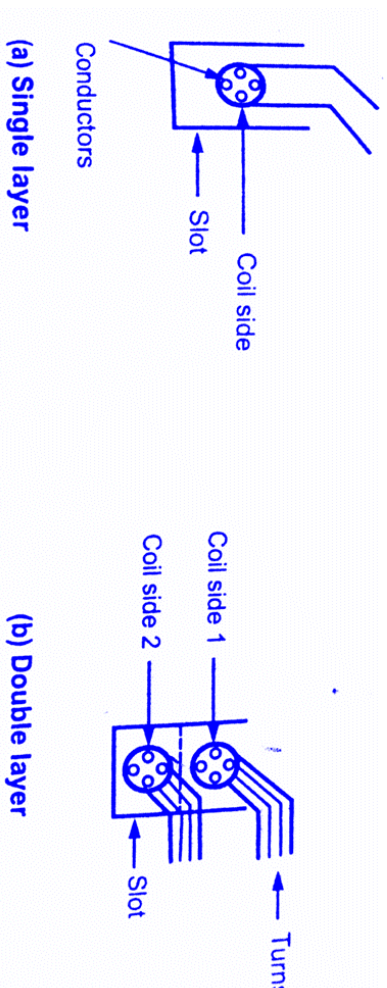
The different types of armature windings in alternators are,

- 1) Single layer and double layer winding
- 2) Full pitch and short pitch winding
- 3) Concentrated and distributed winding

1) Single Layer and Double Layer Winding :

If a slot consists of only one coil side, winding is said to be a single layer.

This is shown in figure(a). While there are two coil sides per slot, one, at the bottom and one at the top the winding is called **double layer** as shown in figure(b). A lot of space gets wasted in single layer hence in practice generally **double layer winding** is preferred.



2) Full Pitch and Short Pitch Winding:

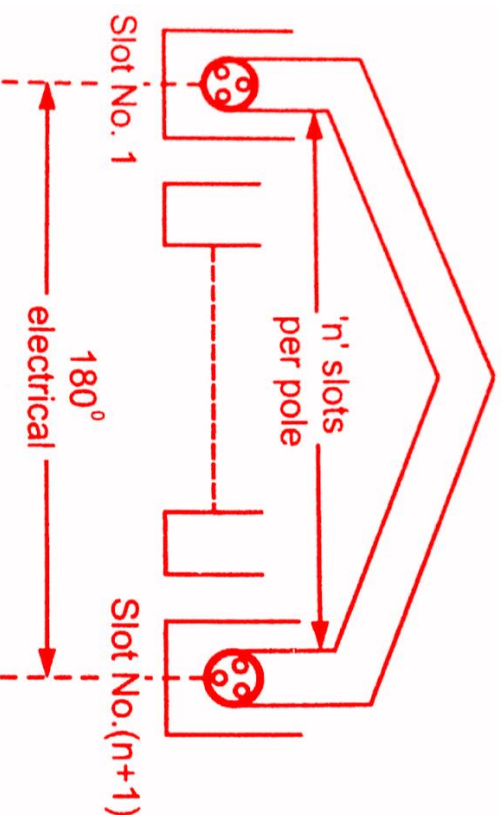
one pole pitch is 180° electrical.

The value of 'n', slots per pole indicates how many slots are contributing 180° electrical phase difference.

So if coil side in one slot is connected to a coil side in another slot which is one pole pitch distance away from the first slot, the winding is said to be **full pitch winding** and coil is called full pitch coil.

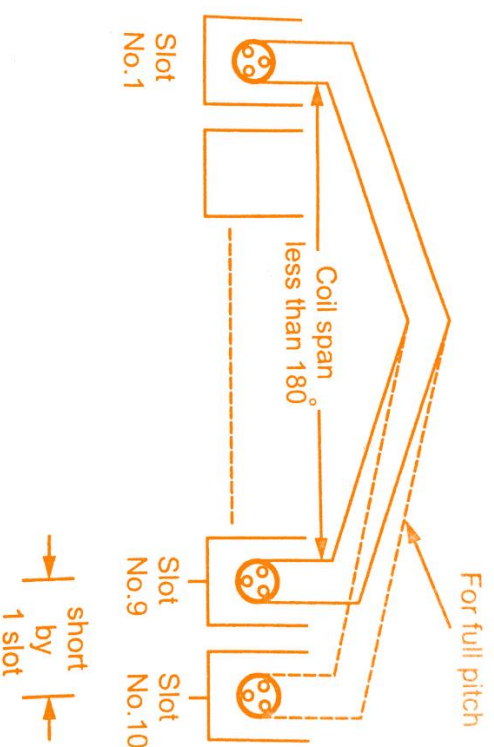
For example, in 2 poles, 18 slots [ktunotes.in](https://www.ktunotes.in) alternator, the pole pitch is $n = 18/2 = 9$ slots. So if coil side in slot No. 1 is connected to coil side in slot No. 10 such that two slots No. 1 and No. 10 are one pole pitch or n slots or 180° electrical apart, the coil is called full pitch coil. Here we can define one more term related to a coil called coil span.

Coil Span:



It is the distance on the periphery of the armature, between two coil sides of a coil. It is usually expressed in terms of number of slots or degrees electrical. So if coil span is 'n' slots or 180° electrical the coil is called 180° full pitch coil. This is shown in the figure to left. As against this if coils are used in such a way that coil span is slightly less than a pole pitch i.e. less than 180° electrical, the coils are called, **short pitched coils** or fractional pitched coils. Generally, coils are shorted by one or two slots.

So in 18 slots, 2 pole alternator instead of connecting a coil side in slot No 1 to slot No.10, it is connected to a coil side in slot No.9 or slot No. 8, the coil is said to be short pitched coil and winding are called **short pitch winding**. This is shown in the below figure.



Advantages of Short Pitch Coils:

In actual practice, short pitch coils are used as it has following advantages,

1) The length required for the end connections of coils is less i.e. the inactive length of winding is less. So less copper is required. Hence economical.

2) Short pitching eliminates high frequency harmonics which distort the sinusoidal nature of e.m.f. Hence waveform of an induced e.m.f. is more sinusoidal due to short pitching.

3) As high frequency harmonics get eliminated, eddy current and hysteresis losses which depend on frequency also get minimised. This increases the efficiency.

3. Concentrated & Distributed winding

Concentrated winding – if there is only one slot/pole/phase (i.e, number of slots = number of poles), then concentrated winding is obtained. Here maximum EMF is induced but is not exactly of sinusoidal form

Distributed winding – If the conductors are placed in several slots under one pole, winding is known as distributed winding.

Advantages of Distributed winding

- certain harmonics can be eliminated
- Armature reaction reduces
- the core is better utilized
- higher current density in copper can be adopted due to even distribution of winding over the surface of armature

4. Integral slot & Fractional slot winding

Integral slot winding – if the number of slots/pole/phase is an integer, the winding is called Integral slot winding
Fractional slot winding – if the number of slots/pole/phase is a fractional number, the winding is called fractional slot winding

$$\text{Slot angle, } \beta = \frac{180^\circ \text{ electrical}}{n}, \text{ where 'n' is the number of slots/pole}$$

Pitch factor, K_p (Chord factor)

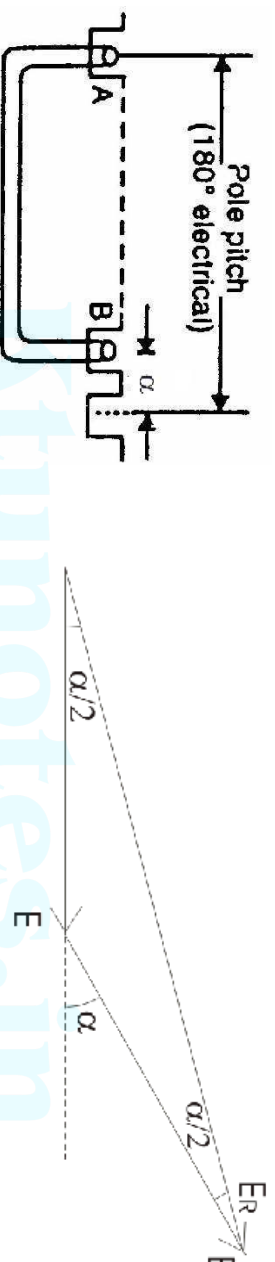
- In a full pitch winding, two coil sides of a coil are under opposite poles, EMF induced in them are equal in magnitude & opposite in phase
- The winding is provided in such a way that these EMFs will add up
- In a short pitch winding the induced EMFs in two sides of the coil are not in phase & hence resultant EMF is given by phasor sum of induced EMFs

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$$\text{Pitch factor, } K_p = \frac{\text{Phasor sum of EMF induced / coil}}{\text{Arithmetic sum of EMF induced / coil}}$$

$$\text{Also, } K_p = \frac{\text{EMF induced in short pitch coil}}{\text{EMF induced in full pitch coil}}$$

Consider a coil AB which is short-pitch by an angle ' α ' electrical degrees (' α ' is always an integer multiple of slot angle ' β '). The EMFs generated in the coil sides A and B differ in phase by an angle ' α ' and can be represented by phasors E_A and E_B respectively as shown in Fig. E_R represents the resultant EMF of the coil.



Since $E_A = E_B = E$, $E_R = 2E \cos \alpha/2$

If the coils are full pitched, then total induced EMF in the coil would have been $= 2E$

$$\text{i.e., } K_p = \frac{2E \cos \alpha/2}{2E} = \cos \alpha/2 \quad \dots\dots (1)$$

Pitch factor is the factor by which the induced EMF in the coil is reduced due to short pitching of coil

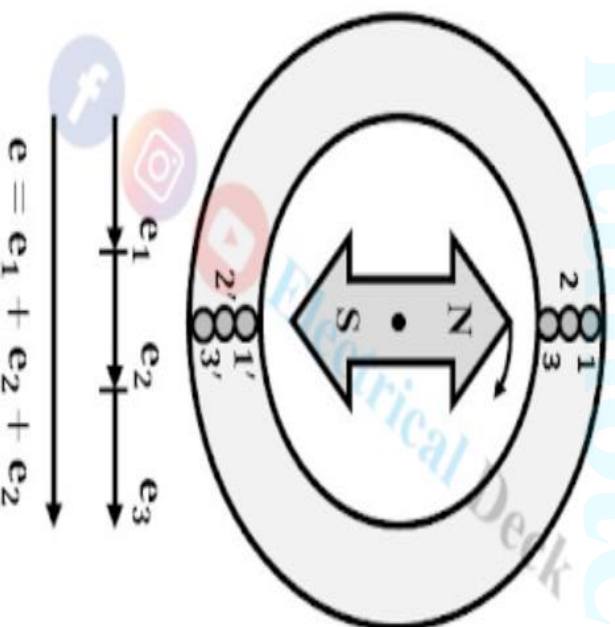
For full pitch winding, $K_p = 1$, for short pitch winding, $K_p < 1$

Harmonic Elimination

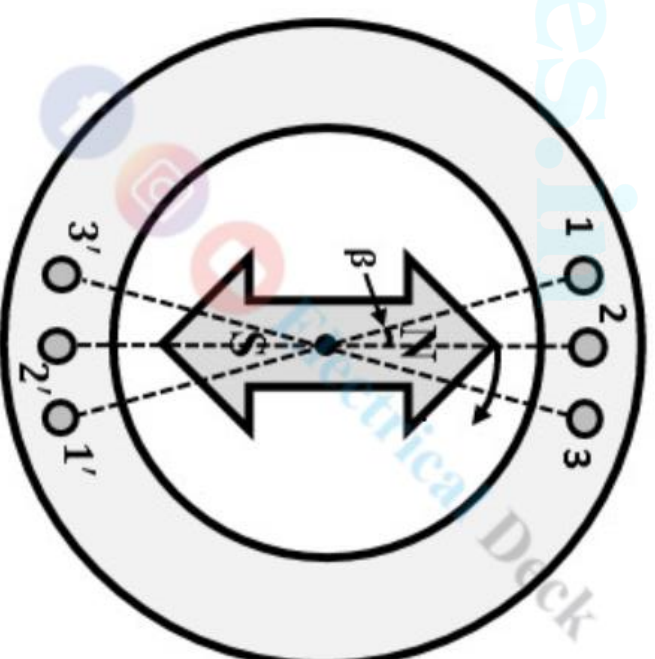
- The equation (1) is the Pitch factor for fundamental component of EMF
- The Pitch factor for r^{th} harmonic EMF is given by , $K_{pr} = \cos(r\alpha/2)$
- The r^{th} harmonic EMF will be reduced to zero if the angle α is made such that $\cos(r\alpha/2) = 0$ or $(r\alpha/2) = 90^\circ$
- 3 phase alternators are star connected to suppress 3^{rd} & 9^{th} harmonics. Suitable short pitching of winding is done to suppress 5^{th} & 7^{th} harmonic components in induced EMF

Distribution factor, K_d (Breadth factor)

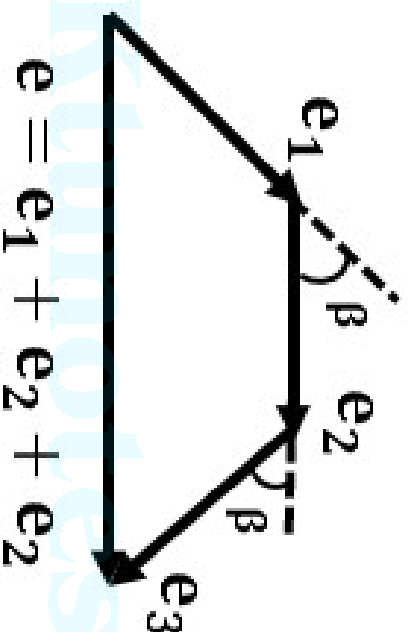
- In concentrated winding, all the coil sides of any one phase are connected in series and grouped together in one slot, and placed under one pole.
- The magnitude and phase difference of the emf induced remains the same throughout the coils of one phase under one pole.
- The arithmetic sum of the emfs induced in each coil gives net emf induced under one pole as shown in the below figure.



- But, in order to obtain a better wave shape (i.e., more sinusoidal of the induced emf) the coils are divided into a number of slots and distributed one each in three slots per-phase under each pole.
- The below figure shows the coils of one phase distributed in adjacent slots, and are connected in series.
- Though the magnitude of the induced emf in distributed winding remains the same, there exists a time-phase difference by a certain angle β° between the coils of one phase under one pole.



The coils are displaced because if we consider the clockwise direction of the alternator as shown above. A similar amount of emf induced in the coil 2-2' as of 1-1' will occur after a time lag of β . Similarly for the coil 3-3' with a time lag of 2β respectively. Hence, the time-phase difference of the induced emf in coils of one phase under one pole depends upon the angular displacement of the slots.



If we represent individual emf induced in coils of the same phase under one pole by phasors. The resultant emf will be phasor sum of all of them as shown above. Here we can see that instead of arithmetic sum for resultant emf as seen in the concentrated winding. It will be a phasor sum for distributed winding. Thus the magnitude of total emf induced will be lesser compared to concentrated type winding.

Distribution factor, K_d (Breadth factor)

- In concentrated winding, the EMF generated/phase is equal to the arithmetic sum of the individual coil EMFs in that phase
- If the winding is distributed type, the EMFs in the coils are not in phase (i.e., phase difference is not zero) but are displaced from each by the slot angle
- The EMF/phase will be the phasor sum of coil EMFs.

- Distribution factor, $K_d = \frac{\text{Phasor sum of EMF induced / coil}}{\text{Arithmetic sum of EMF induced / coil}}$

- Also, $K_d = \frac{\text{EMF induced in distributed winding}}{\text{EMF induced in concentrated winding}}$

Let number of slots/pole = n

Number of slots/pole/phase = m

Induced EMF in each coil side = E

Angular displacement between the slots = slot angle, $\beta = 180/n$

- The different EMFs induced in different coils of one phase under one pole are represented by side AC, CD, DE, EF,... which are equal in magnitude (say E) & are differ in phase by $\square$
- If bisectors are drawn on AC, CD, DE, EF,... they would meet at common point at 'O'.
- EMF induced in each coil side, $E = AC = 2 * OA \sin(\beta / 2)$
- Arithmetic sum of EMF induced

$$= m * 2 * OA \sin(\beta / 2)$$
- Phasor sum of EMF induced

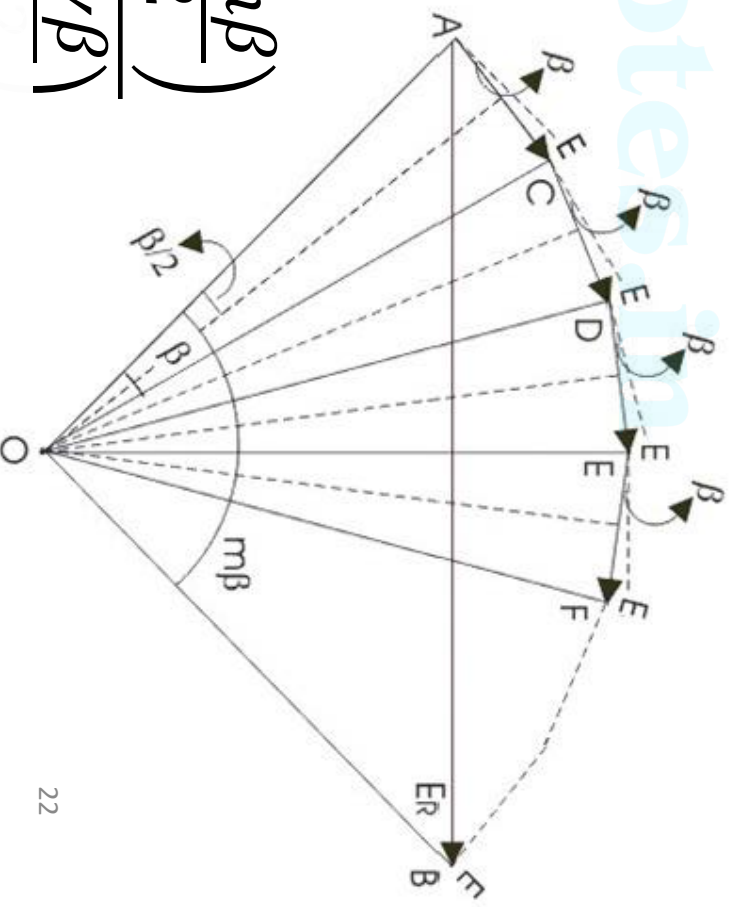
$$= E_R = 2 * OA \sin(<AOB/2)$$

$$= 2 * OA \sin(m\beta/2)$$

i.e, $K_d = \frac{\sin\left(\frac{m\beta}{2}\right)}{m \sin\left(\frac{\beta}{2}\right)}$

$$\sin\left(\frac{\gamma m \beta}{2}\right)$$

For r^{th} harmonic, $K_{dr} = \frac{\sin\left(\frac{\gamma m \beta}{2}\right)}{m \sin\left(\frac{\gamma \beta}{2}\right)}$



EMF equation

Let, Z = number of conductors in series/phase

ϕ = flux/pole in Wb

P = Number of rotor poles

N = Rotor speed in rpm

Time for one revolution, $t = 60/N$ sec

Flux cut by the stator conductor in one revolution, $d\phi = P\phi$ Wb

Average EMF induced in one stator conductor = $d\phi/dt$

$$= \frac{P\phi}{60/N} = \frac{P\phi N}{60} \text{ Volts}$$

Since there are Z conductors in series/phase, Average EMF/phase

$$= \frac{P\phi N}{60} * Z$$

$$= \frac{P\phi Z}{60} * \frac{120f}{P} = 2f\phi Z \text{ Volts}$$

$$N = \frac{120f}{P}$$

- RMS value of EMF/phase = Average value/phase * form factor

$$= 2f\phi Z * 1.11 = 2.22 f\phi Z \text{ Volts}$$
- If K_p & K_d are the Pitch factor & Distribution factor of armature winding, then,

$$E_{\text{RMS/phase}} = 2.22 K_p K_d f\phi Z \text{ Volts}$$

- Sometimes turns/phase (T) is given rather than conductors/phase (Z = 2T). Then,

$$E_{\text{RMS/phase}} = 4.44 K_p K_d f\phi T \text{ Volts}$$

The product of pitch factor & distribution factor is called winding

$$\text{factor } K_w = K_p * K_d$$

Harmonics in output voltage waveform

- Harmonics are signals having a frequency which is the integer multiple of fundamental frequency
- Source of harmonics in output voltage waveform are
 - Non sinusoidal waveform of field flux
 - Variation in reluctance of air gap due to slotting of stator core

- In a salient pole machine, the air gap is made to vary sinusoidally around the machine by proper construction of field poles
- In a non-salient pole machine, the air gap is uniform. Therefore to get a field of sinusoidal form, the field windings are distributed in slots sinusoidally
- An ideal sinusoidal field form is very difficult to achieve and therefore harmonic EMFs will be induced in winding
- The harmonics can be eliminated from output voltage waveform by properly designing the stator windings
- The various methods for elimination of harmonics are
 - Distributing the armature winding
 - Short pitching the armature winding
 - By using fractional slot winding
 - larger air gap length causes an increase in reluctance & harmonics will reduce

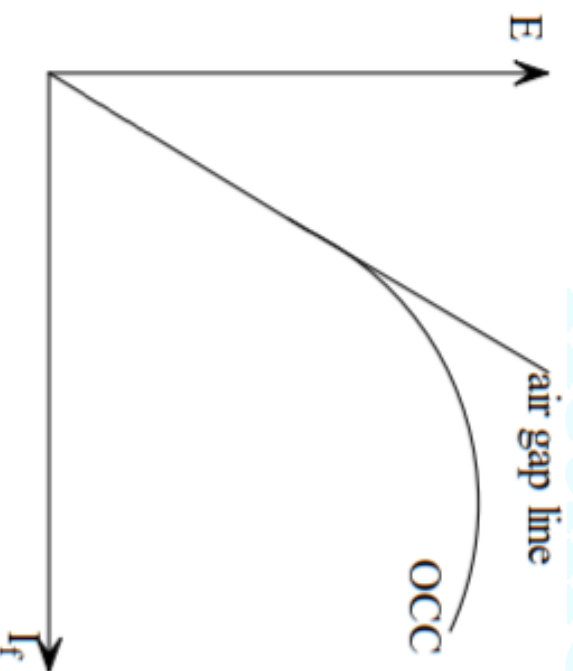
Synchronous generator on no load

When a synchronous generator is driven at constant speed, the terminal voltage on open circuit is,

$$E = K \cdot \Phi \quad \text{where} \quad K = 4K_f K_e K_d f T$$

$$\phi \propto I_f$$

Therefore, $E \propto I_f$



Synchronous generator on load

- When a synchronous generator is running on no-load, there will be no current flowing through the armature windings. The flux produced in the air-gap will be due to the field ampere-turns only.
- When load is connected across the armature terminals, current will flow through the armature windings.
- This three phase currents will produce a rotating magnetic field in the air-gap.
- The effect of the armature flux on the flux produced by the field ampere-turns is called armature reaction.
- The armature flux will distort, oppose or aid the field flux causing reduction or increase in the air-gap flux depending upon the power factor of the load.

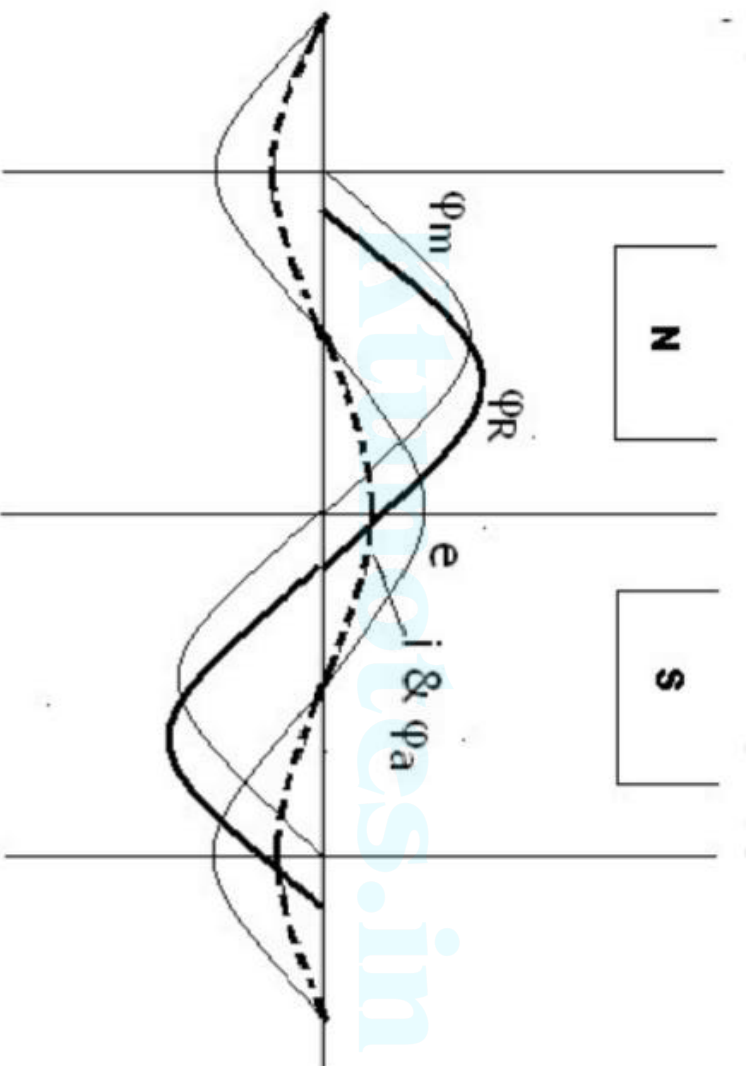
ARMATURE REACTION

- The effect of the armature flux (produced by the armature current) on the main flux (produced by the field winding) is called armature reaction.
- The effect of the armature flux depends on the magnitude of the armature current and load power factor.

Main flux ϕ_m available in the air-gap is sinusoidal. Sinusoidal main flux produces a sinusoidal emf and hence a sinusoidal current. Sinusoidal armature current causes the armature flux also to be sinusoidal. Armature flux ϕ_a and main flux ϕ_m are rotating at the same speed in same direction. Hence the relative speed between ϕ_a and ϕ_m is zero or they are stationary in space.

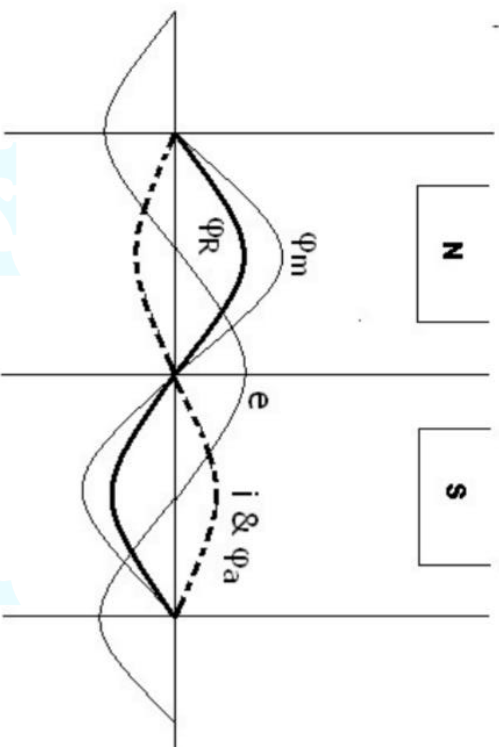
1. Armature reaction at unity power factor

Consider a purely resistive load connected to the alternator having unity power factor.



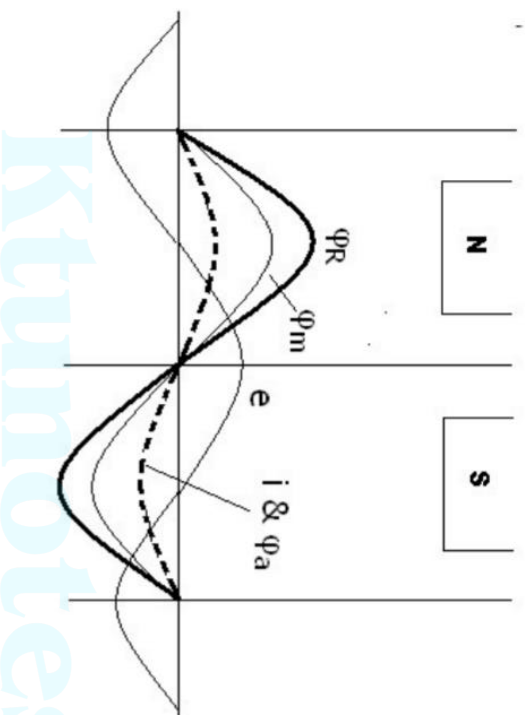
- Induced emf e lags behind ϕ_m by 90° ($e \propto -d\phi/dt$).
- Since power factor is unity, e and i will be in phase.
- i sets up a flux ϕ_a in phase with i .
- Resultant flux in the air-gap ϕ_R will be the sum of ϕ_a and ϕ_m .
- Here, ϕ_a lags behind ϕ_m by 90° .
- Due to armature reaction, flux in the first quarter portion is decreased and in the second quarter portion flux is increased.
 - If saturation is neglected, amount of flux increased in the first quarter and amount of flux decreased in the second quarter will be equal. Hence, the effect of armature reaction is only distortional or cross-magnetizing at unity power factor.
 - If saturation is accounted (i.e. if the armature current is large), amount of flux increased in the second quarter will be less than the amount of flux decreased in the first quarter and hence the resultant flux ϕ_R will be less than ϕ_m in magnitude.
- Hence, in a saturated machine, the effect of armature reaction is both distortional and demagnetizing.

2. Armature reaction at zero power factor lagging



- Induced emf e lags behind ϕ_m by 90° .
- Since power factor is zero lagging, i lags behind e by an angle 90° .
- i sets up a flux ϕ_a in phase with i . ϕ_a lags behind ϕ_m by 180° or ϕ_a is in phase opposition to ϕ_m .
- Resultant flux in the air-gap ϕ_R will be the sum of ϕ_a and ϕ_m . Resultant flux ϕ_R is decreased.
- Effect of armature reaction is purely demagnetizing for a purely inductive load (Zero power factor lagging).
- Since the resultant flux is decreased, generated emf will be decreased.

3. Armature reaction at zero power factor leading



- Induced emf e lags behind ϕ_m by 90° .
- Since power factor is zero leading, i leads from e by an angle 90° .
- i sets up a flux ϕ_a in phase with i . ϕ_a is in phase with ϕ_m .
- Resultant flux in the air-gap ϕ_R is increased.
- Effect of armature reaction is purely magnetizing for a purely capacitive load (Zero power factor leading).
- Since the resultant flux is increased, generated emf will be increased.

For an R-L load ($0 < \text{p.f. lag} < 1$), effect of armature reaction is partly distortional and partly demagnetizing.

For an R-C load ($0 < \text{p.f. lead} < 1$), effect of armature reaction is partly distortional and partly magnetizing.

The change in terminal voltage due to effect of armature reaction can be accounted by assuming the presence of a fictitious reactance X_{ar} in series with the armature

Armature leakage reactance

- When current flows in the armature (i.e. when the machine is loaded), it sets up armature flux.
- Some part of the armature flux links with the main flux causes armature reaction. The remaining part is called leakage flux.

The armature flux which links only with the armature winding not with the field winding is called leakage flux.

The various

leakage fluxes may be

- I. slot leakage flux – This includes the fluxes that link the coils of a winding by crossing the slots in which the coils are placed
 - II. overhang leakage flux - includes all the leakage fluxes around the coil ends protruding from the slots at each end of the core.
 - III. zig-zag leakage flux - is due to fluxes produced by a given winding that cross the air gap but do not link the other winding.
- The effect of this leakage flux is accounted by connecting a leakage reactance X_{al} in series with the armature.

Synchronous impedance

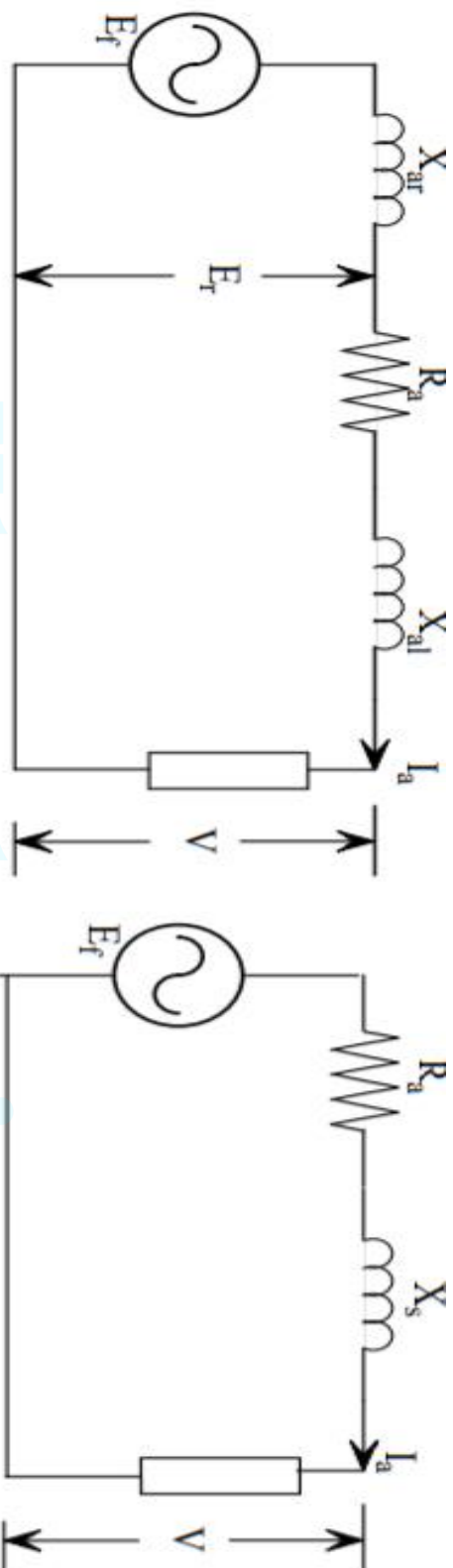
When a synchronous generator is loaded, there will be a change in the terminal voltage from that under no-load condition, due to :

- i) armature resistance R_a
- ii) armature leakage reactance X_{al}
- iii) effect of armature reaction X_{ar}

$$V = E_f - I_a R_a - jI_a X_{al} - jI_a X_{ar} = E_f - I_a R_a - jI_a (X_{al} + X_{ar})$$

$$\text{or } E_f = V + I_a R_a + jI_a (X_{al} + X_{ar}) = V + I_a R_a + jI_a X_s$$

$$\text{where } X_s = X_{al} + X_{ar} = \text{Synchronous reactance}$$



$Z_s = R_a + jX_s$ = synchronous impedance

$\bar{\phi}_r = \bar{\phi}_f + \bar{\phi}_{ar}$

$\bar{\phi}_r$ = air gap flux (resultant flux)

$\bar{\phi}_f$ = main flux

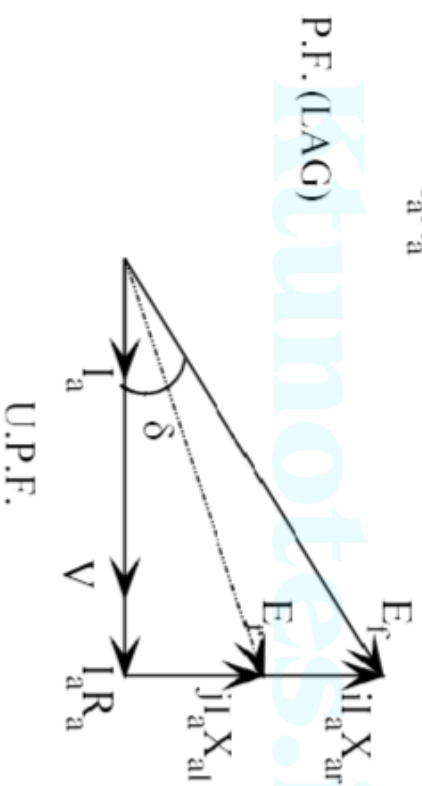
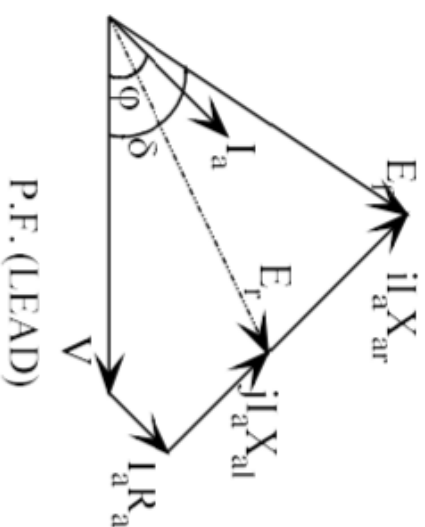
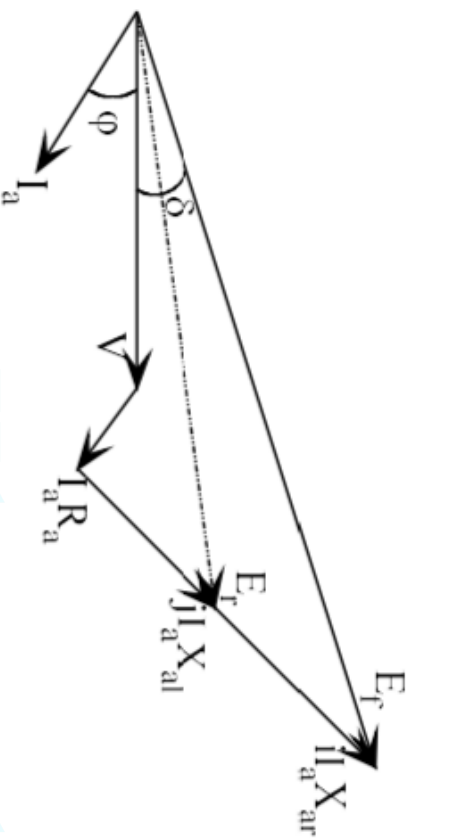
$\bar{\phi}_{ar}$ = armature flux

E_f = air gap emf

E_f = excitation emf

V = terminal voltage

PHASOR DIAGRAMS

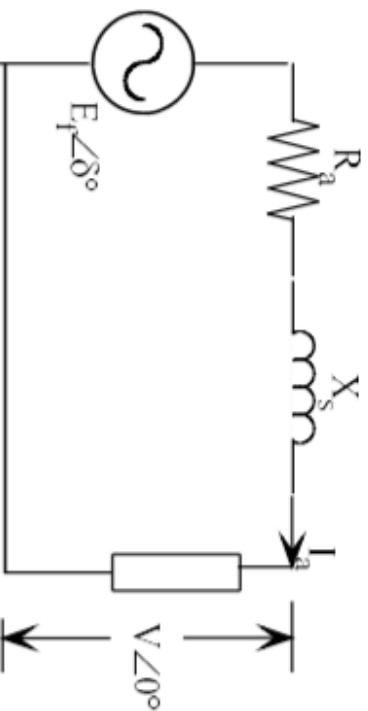


δ = load angle = angle between E_f and V

ϕ = power factor angle = angle between V and I_a

θ = synchronous impedance angle = $\tan^{-1} \frac{X_s}{R_a}$

POWER FLOW EQUATIONS (Synchronous Generator)



V = Terminal voltage/phase

E_f = Excitation voltage/phase

I_a = Armature current

δ = Load or power angle (angle between E_f and V)

θ = Synchronous impedance angle = $\tan^{-1}\left(\frac{X_s}{R_a}\right)$ (if R_a is neglected, $\theta = 90^\circ$)

ϕ = Power factor angle = angle between V and I_a .

$\bar{V} = V \angle 0^\circ$, $\bar{E}_f = E_f \angle \delta^\circ$

Synchronous impedance = $R_a + jX_s = Z_s \angle \theta$

Complex Power Output per phase

$$S_o = P_o + jQ_o = \bar{V} \times \bar{I}_a^*$$

$$= V \angle 0^\circ \times \left(\frac{E_f \angle \delta - V \angle 0^\circ}{Z_s \angle \theta} \right)^*$$

$$= V \angle 0^\circ \times \left(\frac{E_f}{Z_s} \angle (\delta - \theta) - \frac{V}{Z_s} \angle -\theta \right)^*$$

$$= \frac{VE_f}{Z_s} \angle (\theta - \delta) - \frac{V^2}{Z_s} \angle \theta$$

$$\text{Real power output per phase, } P_o = \frac{VE_f}{Z_s} \cos(\theta - \delta) - \frac{V^2}{Z_s} \cos \theta = \frac{VE_f}{Z_s} \cos(\theta - \delta) - \frac{V^2 R_a}{Z_s^2}$$

$$= \frac{VE_f}{Z_s} \cos(\theta - \delta) - \frac{V^2 R_a}{Z_s^2} \text{ since } \cos \theta = \frac{R_a}{Z_s}$$

= Electrical power developed by the generator

$$\text{Reactive power output per phase } Q_o = \frac{VE_f}{Z_s} \sin(\theta - \delta) - \frac{V^2}{Z_s} \sin \theta$$

Complex Power Input per phase

$$S_i = P_i + jQ_i = \bar{E}_f \times \bar{I}_a^*$$

$$= E_f \angle \delta \times \left(\frac{E_f \angle \delta - V \angle 0}{Z_s \angle \theta} \right)^*$$

$$= E_f \angle \delta \times \left(\frac{E_f}{Z_s} \angle (\delta - \theta) - \frac{V}{Z_s} \angle -\theta \right)^*$$

$$= \frac{E_f^2}{Z_s} \angle \theta - \frac{VE_f}{Z_s} \angle (\theta + \delta)$$

$$\text{Real power input per phase, } P_i = \frac{E_f^2}{Z_s} \cos \theta - \frac{VE_f}{Z_s} \cos(\theta + \delta)$$

$$\text{Reactive power input per phase } Q_i = \frac{E_f^2}{Z_s} \sin \theta - \frac{VE_f}{Z_s} \sin(\theta + \delta)$$

Mechanical power input to generator = P_i + rotational losses

Rotational losses = Core loss + Mechanical loss

Core loss = Eddy current loss + Hysteresis loss

Mechanical loss = Friction loss + Windage loss

Maximum Power Output per phase

For maximum power output, $\frac{dP_o}{d\delta} = 0$

$$\frac{d}{d\delta} \left(\frac{VE_f}{Z_s} \cos(\theta - \delta) - \frac{V^2 R_a}{Z_s^2} \right) = 0$$

Since V , E_f , Z_s and R_a are constants,

$$\sin(\theta - \delta) = 0 \text{ or } \delta = \theta$$

Maximum power output occurs if load angle = impedance angle

$$\text{Maximum power output, } P_{o \max} = \frac{VE_f}{Z_s} - \frac{V^2 R_a}{Z_s^2}$$

If R_a is neglected, for maximum output, $\delta = 90^\circ$ and $P_{o \max} = \frac{VE_f}{X_s}$

Maximum Power Input per phase

For maximum power input, $\frac{dP_i}{d\delta} = 0$

$$\frac{d}{d\delta} \left(\frac{E_f^2}{Z_s} \cos \theta - \frac{VE_f}{Z_s} \cos(\theta + \delta) \right) = 0$$

Since V , E_f , Z_s and θ are constants,

$$\sin(\theta + \delta) = 0 \text{ or } \theta + \delta = 180^\circ$$

For maximum power input, $\delta = 180 - \theta$.

$$\text{Maximum power input} = P_{i\max} = \frac{E_f^2}{Z_s} \cos \theta + \frac{VE_f}{Z_s}$$

$$\text{If } R_a \text{ is neglected, } \delta = 90^\circ, \quad P_{i\max} = \frac{VE_f}{X_s}.$$

LOSSES & EFFICIENCY

The various losses in synchronous machines are

- I. No-load rotational losses (friction and windage loss, open-circuit core loss)
- II. Field circuit loss ($I^2 R$ loss in the field winding)
- III. Direct load loss ($I^2 R$ loss in the armature winding)
- IV. Stray load losses (core loss in the teeth, core end-plates etc due to armature leakage flux, eddy current loss in the armature conductors due to armature leakage flux)

Direct load loss + stray load loss = short circuit load loss